

S O T E R I A

DESIGN PURPOSE



This design aims to improve the awareness of drivers and cyclists in a potential intersection collision.

DESIGN VALUES AND DFX

SAFETY

Detect cars up to 10m
Alerts user with sounds and lights

ACCESSIBILITY

Mountable on any bike
Quick to implement

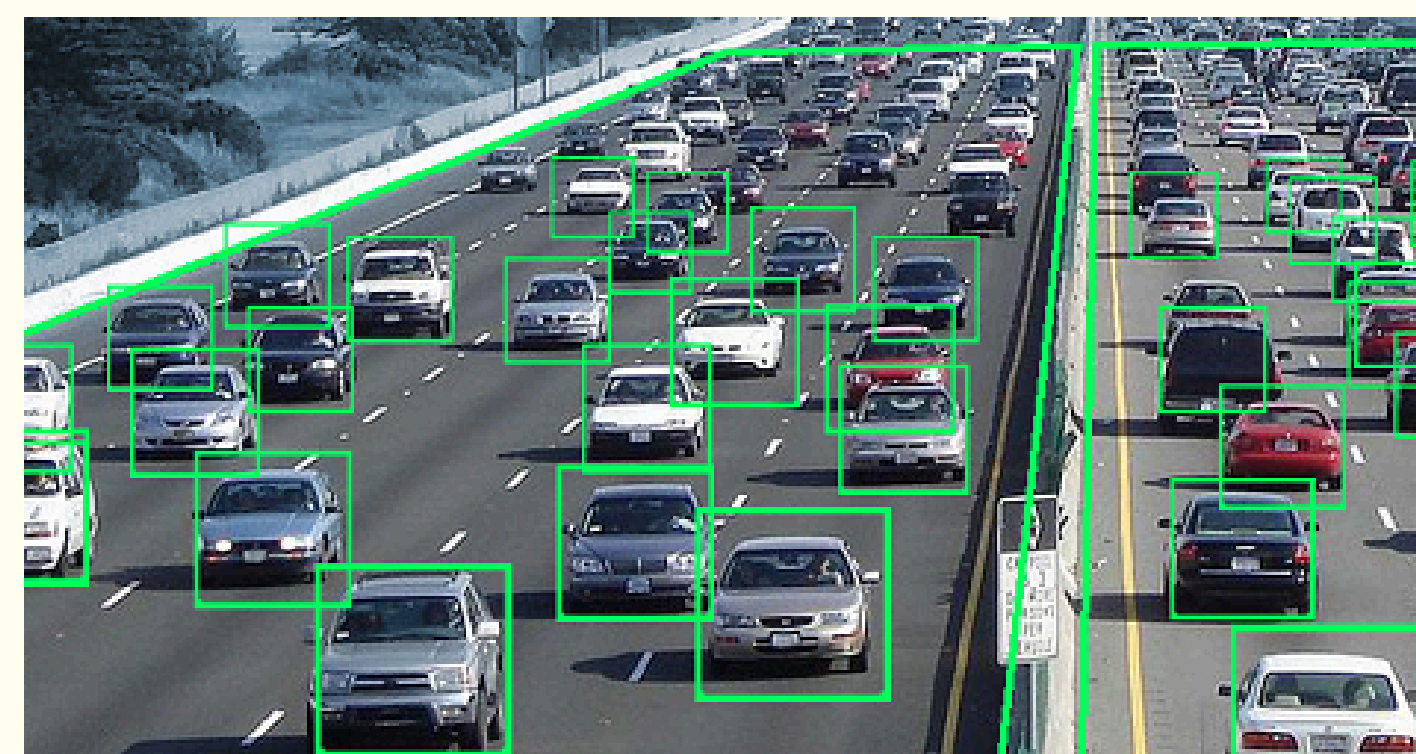
CONVENIENCE

Automatic GPS and computer vision
Adjustable phone and camera mount

1

COMPUTER VISION

Webcam monitors vehicles in blind-spot with computer vision



1



4



2

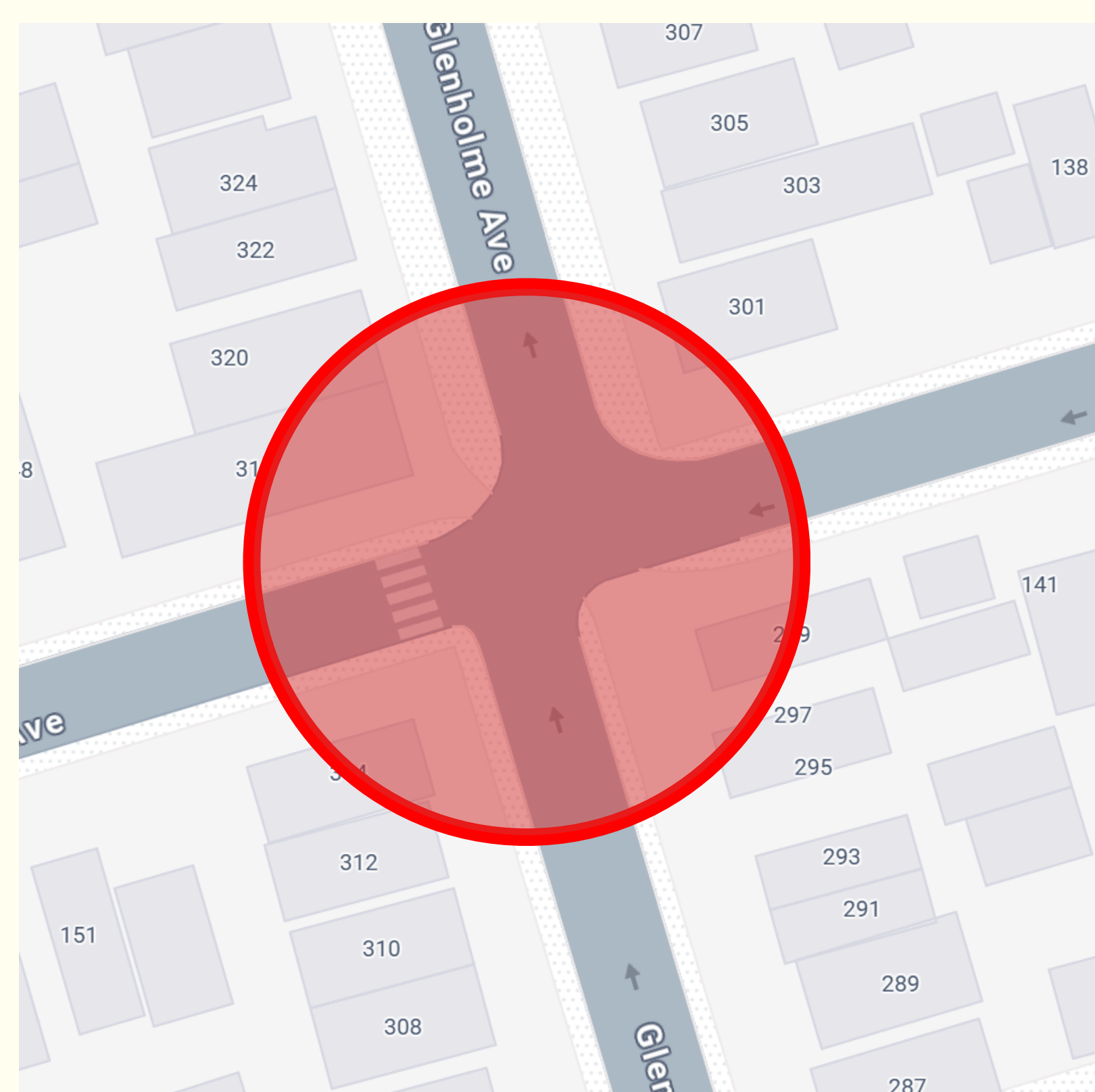
3



2

GPS SYSTEM

Software detects position at intersection, and activates computer when cars are detected



3

ALERT SYSTEM

When in intersection and car detected, the phone vibrates and lights up



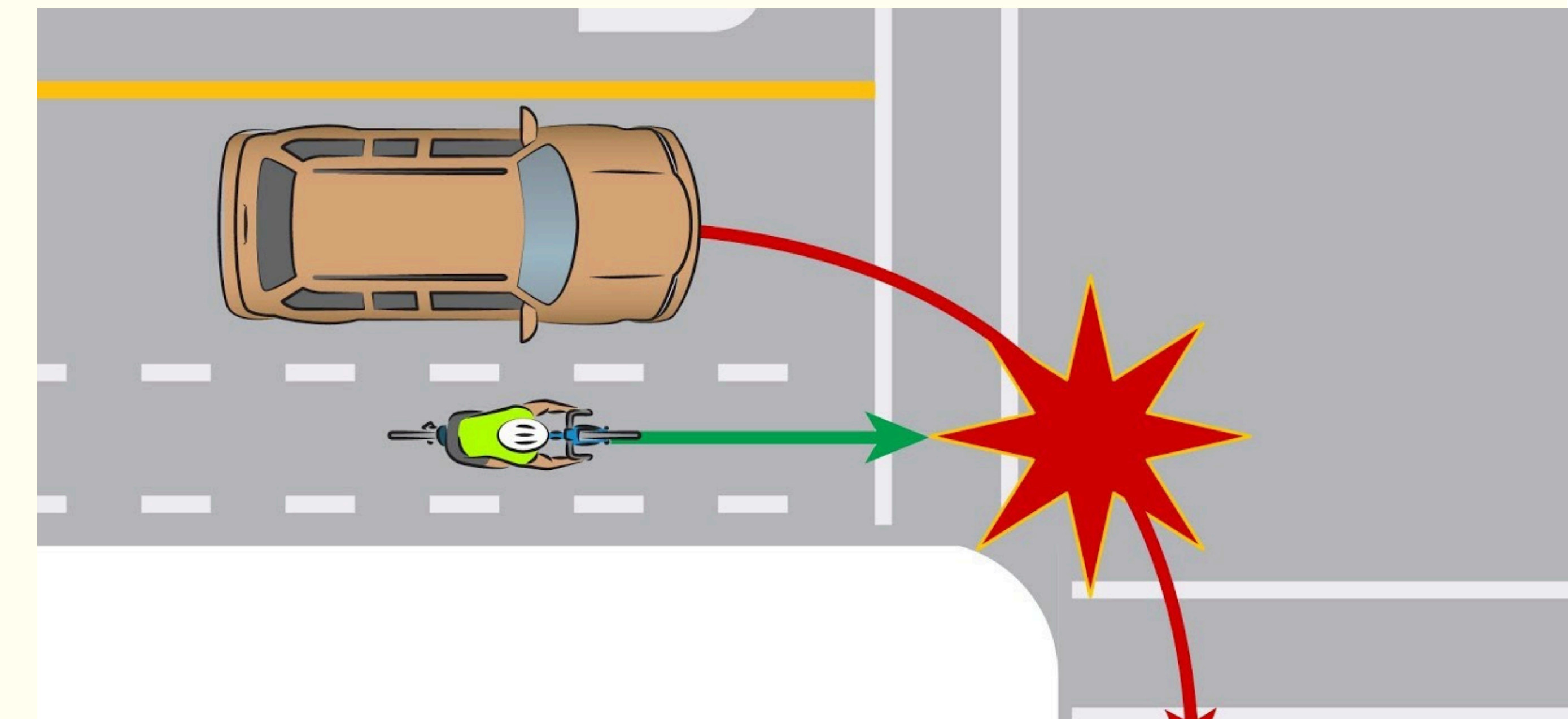
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ELECTRIC HORN

User can choose to respond with a honk using the electric horn



RIGHT HOOK COLLISIONS



Occurs when a car in a biker's blind-spot turns right at an intersection while the biker travels forwards.

NEXT STEPS

WIRELESS CAMERA

Weather and dust-proof
Reduce hazards from tangling wires
Reduce load on phone battery

EXTERNAL ALERTS

Signals to external light and phone
Other haptic devices for greater accessibility (vibrating bracelet)
Reduce phone battery drain